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Roll No

EE/EX-5003-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Switchgear and Protection

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the different types of fault occur in power system?
b) State the necessity of current limiting reactors in power system and classify the reactor on the basis of their classification.
2. a) Discuss the merits of per unit computation in power system.
b) With the help of sequence network and relevant mathematical expression, derive the expression for fault current on a solidly grounded unloaded alternator. When subjected to single line to ground fault.
3. a) Discuss the functional characteristics of a protection relay and define the following:
 - i) Pickup value
 - ii) Reset value
 - iii) P.S.M
 - iv) IDMT characteristics.

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- b) With the help of neat diagram explain operation of induction type over current relay.
- 4. a) Explain the neat diagram the working and operation of distance relay.
b) What is meant by directional features of directional relay? Describe the construction, principle of operation and application of a directional over current relay.
- 5. a) What is RRRV? State the formula of restriking voltage.
b) What are the different methods of arc interruption.
- 6. Explain the working principle, construction and merits of minimum oil CB with the help of neat sketch.
- 7. a) State the abnormalities and faults in alternator with necessary protection.
b) Explain in detail percentage differential protection scheme of transformer.
- 8. a) What do you mean by insulation co-ordination and give the significance of basic insulation levels in insulation co-ordination
b) Explain the phenomena of lightning and protection provided against lightning.

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